

CUSTOMER SEGMENTATION & CHURN ANALYSIS REPORT Dataset:

Kaggle Bank Customer Churn Dataset Records Analysed: 10,000 |

Tools: Python, pandas, scikit-learn, matplotlib

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EXECUTIVE SUMMARY

Analysis of 10,000 bank customer records revealed an overall churn rate of 20.4%. Using K-Means clustering, customers were segmented into 4 distinct behavioural profiles. A Random Forest model was trained to predict individual churn probability, achieving 86.4% accuracy and an AUC of 0.85 — significantly outperforming the Logistic Regression baseline (AUC 0.77).

SEGMENT PROFILES

Segment 0 — Low-Balance Actives (25.6% of base, 2,562 customers)

Avg Balance: £4,494 | Active Members: 52.5% | Churn Rate: 15.7%

→ Strategy: Upsell to savings products. Low balance suggests untapped deposit potential.

Segment 1 — Premium Engaged (30.7% of base, 3,074 customers)

Avg Balance: £121,368 | Active Members: 100% | Churn Rate: 15.7%

→ Strategy: Maintain engagement through loyalty rewards. Lowest-risk, highest-value retained segment.

Segment 2 — At-Risk Mid-Tier (15.0% of base, 1,497 customers)

Avg Balance: £19,135 | Active Members: 48.8% | Churn Rate: 20.2%

→ Strategy: Re-engagement campaign targeting inactive members in this group.

Segment 3 — High-Value Disengaged (28.7% of base, 2,867

customers) Avg Balance: £122,642 | Active Members: 0% | Churn

Rate: 29.7% → Strategy: URGENT. Highest priority for retention. Zero active members despite highest average balance.

KEY FINDINGS

Finding 1 — Germany Has a Critical Retention Problem Germany shows 32.4% churn — exactly double France (16.2%) and Spain (16.7%). This is not a product issue — it is a market-specific problem requiring a dedicated regional strategy. → Recommendation: Launch a Germany-specific retention campaign. Investigate service quality and competitor activity in that market.

Finding 2 — The Product Paradox Customers with 2 products churn at only 7.6%. Customers with 3 products churn at 82.7%. Customers with 4 products churn at 100% — every single one left. → Recommendation: Cap cross-selling at 2 products per customer. Conduct exit interviews with 3-4 product customers to identify dissatisfaction triggers.

Finding 3 — 1,774 High-Risk Customers Identified The Random Forest model flagged 1,774 customers (17.7% of base) as High Risk. Segment 3 carries the highest average churn probability at 29.7% — and holds an average balance of £122,642. → Recommendation: Trigger personalised outreach for all 1,774 High-Risk customers within 30 days. Prioritise Segment 3 first given their balance value.

Finding 4 — Age is the Strongest Churn Predictor Age ranked as the top feature in the Random Forest model, followed by EstimatedSalary and CreditScore. Churned customers are concentrated in the 40–60 age bracket. → Recommendation: Design targeted retention products for the 40–60 age group — fixed-term deposits, wealth management services, or premium account tiers.

MODEL PERFORMANCE Logistic Regression: AUC = 0.77 Random Forest: 86.4% accuracy, AUC = 0.85 Top 3 Predictors: Age, EstimatedSalary, CreditScore